Prerequisites

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Prerequisite that you need for a successful setup and installation of IBM Fusion.

Procedure

1. Go through the system requirements and confirm whether you meet all the requirements. For detailed information about system requirements, see System requirements.
2. For the prerequisites of individual IBM Fusion services, go through their respective *Before you begin* section.
   For information about the services, see Fusion services.
3. Ensure that you have the infrastructure and supported version of Red Hat OpenShift Container Platform.

   > **Note:**
   > - You can use Red Hat OpenShift Kubernetes Engine (OKE) as an alternative for OpenShift Container Platform. The supported version of OKE is same as OpenShift Container Platform.
   > - IBM Fusion provides support for Red Hat OpenShift Virtualization Engine (OVE) as an alternative for OpenShift Container Platform. If you do not need all the capabilities of OpenShift for initial migrations of VMs and want a simplified and less costly alternative edition of self-managed OpenShift, then you can use OVE for running VMs. Consider the following inclusions and restrictions in the entitlement before you proceed with OVE:
   >   - OVE restricts your application instances to run as a VM only. RHEL entitlements for the VMs must be purchased separately if needed.
   >   - End user container workloads in OVE can only run on independent clusters deployed on the virtual machines which must be provided with separate core-based subscriptions for OpenShift.
   >   - Infrastructure workloads such as storage drivers, backup applications, forwarding agents, Red Hat Advanced Cluster Management for Kubernetes, and Ansible Automation Platform, are permitted to run as containers in OVE.
   >   - Containers that are qualified to run on infrastructure nodes according to the self-managed Red Hat OpenShift subscription guide can also be run on infrastructure nodes in an OVE cluster.
   >   - Fusion Data Foundation including MCG, Scale CNSA, Backup & Restore Spoke, Fusion Control Tower are permitted to run on infrastructure nodes.
   >   - IBM Data Cataloging, Backup & Restore Hub and Fusion Base can be deployed as containers on OVE but must run on normal worker nodes.

   For the supported versions of OpenShift Container Platform, see Support matrix.

> **Note:** If you want to install Data Foundation storage in consumer mode, then OpenShift Container Platform must also be the same version.

4. Obtain IBM Entitlement Registry Key.
   For the steps to obtain the key, see Obtaining entitlement key.
5. Create a pull secret.
   For the actual steps to generate, see Creating image pull secret.
   For actual steps to create a pull secret for installing IBM Fusion on IBM cloud, see Creating image pull secret for IBM Cloud based installation.
6. If you plan to deploy IBM Fusion on a OpenShift Container Platform cluster with a cluster wide proxy configured, then ensure you add these hosts to your allowlist:
   - registry.redhat.io
   - quay.io
   - cp.icr.io
   - icr.io
   - redhat.com
   - registry.connect.redhat.com
   - catalog.redhat.com
   - access.redhat.com
   - cloud.redhat.com
   - cdn02.quay.io
   - registry.access.redhat.com
   - dd0.icr.io
   - dd2.icr.io
   - dd4.icr.io
   - dd6.icr.io

   > **Note:** This step is applicable only for on-premise deployments.

7. Label GPU nodes to avoid IBM Fusion pods from being scheduled on them. It frees up resources on the GPU nodes to be used by AI workloads.
   For detailed guidance on implementation, see AI workloads on IBM Fusion.

– **Obtaining entitlement key**
  Entitlement keys determine whether IBM Fusion software operator can automatically pull the require container native images. During installation, image pull failures may occur due to an invalid entitlement key or a key belonging to an account that does not have entitlement to IBM Fusion.

– **Creating image pull secret**
  Create an image pull secret to enable the OpenShift cluster to authenticate with the IBM Entitled Registry. The image pull secret provides the credentials for pulling Docker images from the IBM Entitled Registry.

- **Creating image pull secret for IBM Cloud based installation**
  With OpenShift Container Platform, you can create a global image pull secret that each worker node i the cluster can use to pull images from a private registry and you can set up your Red Hat OpenShift o IBM Cloud cluster to pull entitled software.
- **System requirements**
  The system requirements for installing IBM Fusion software.
- **SecurityContextConstraints**
  Red Hat OpenShift `SecurityContextConstraints` requirement controls pod permissions. These permissions include pod actions and resources that it can access.

Parent topic:

→ Deploying IBM Fusion